

Howard Tang

Full-Stack Software Engineer

AI SYSTEMS • DISTRIBUTED BACKEND • PRODUCT ARCHITECTURE

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A quick map of what's inside — five shipped products, the stack behind them, and how I approach building.

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Engineering Stack

The languages, frameworks, and infrastructure behind the projects in this portfolio.



Backend & Core

Node.js

NestJS

Python

FastAPI

C / C++



Frontend

React.js

Next.js

TypeScript



Cloud & Data

AWS

Docker

PostgreSQL

DynamoDB

MongoDB



AI & Systems

LLM Workflows

Distributed Systems

Transaction Design

PROJECT 01

Lumi

AI Travel Booking & Discovery Platform

Project Goal

Architect an AI travel platform integrating LLMs with live OTA inventory and POI discovery.

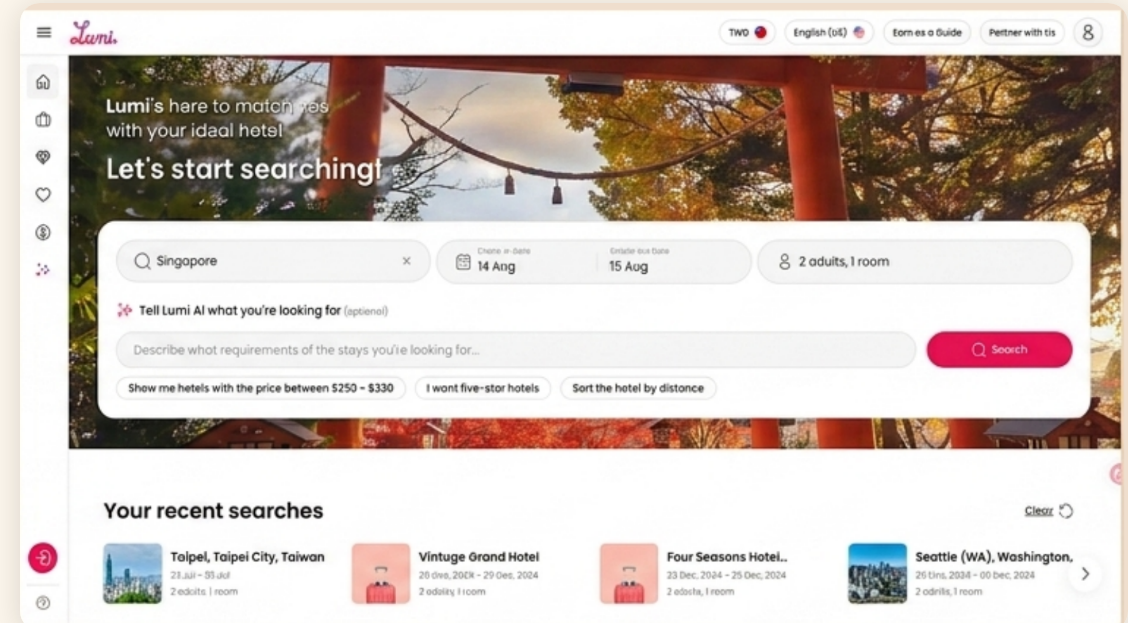
Core Challenge

Isolating **non-deterministic AI generation** from **deterministic booking execution** to ensure transaction reliability.

Impact

Scaled seamlessly to handle **250K+** daily requests globally.

250K+ REQUESTS / DAY



PROJECT 02

FifthKeys

AI-Powered Hotel Property Management System (PMS)

 Project Goal

Design the relational data model and RBAC backend for a real-time hospitality management platform (PMS).

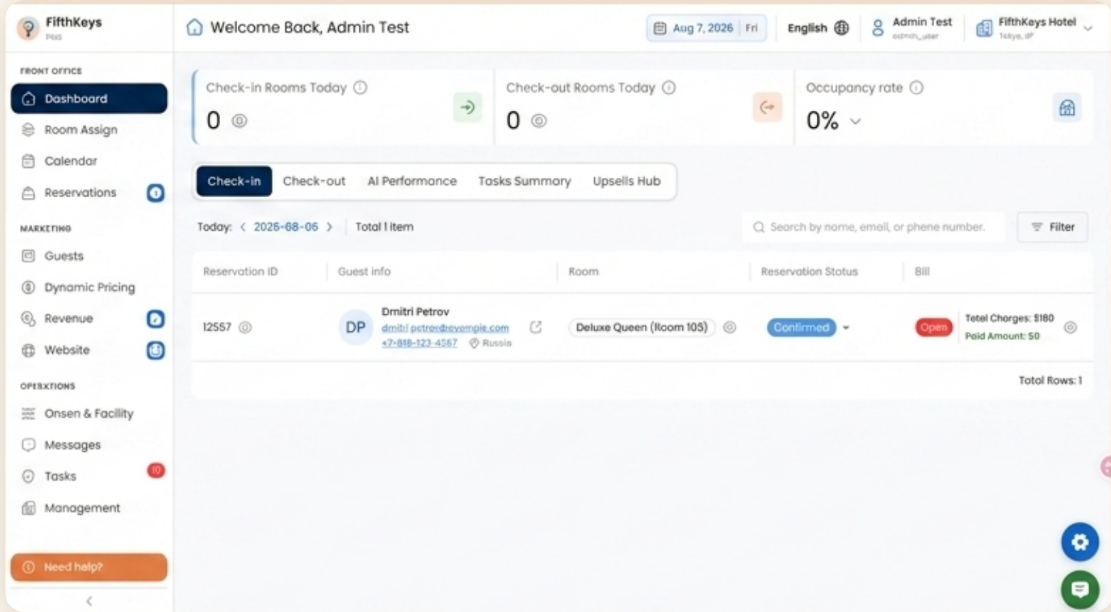
 Core Challenge

Guaranteeing **strict transactional consistency** across inventory availability, reservations, and dynamic pricing.

 Impact

Synchronized real-time channels, directly increasing client hotel occupancy by **20%+**.

+20% OCCUPANCY RATE



PROJECT 03

MixerBox ChatAI

Multi-Modal AI Assistant Platform

Project Goal

Build the backend workflow orchestration and APIs for a production-grade AI assistant.

Core Challenge

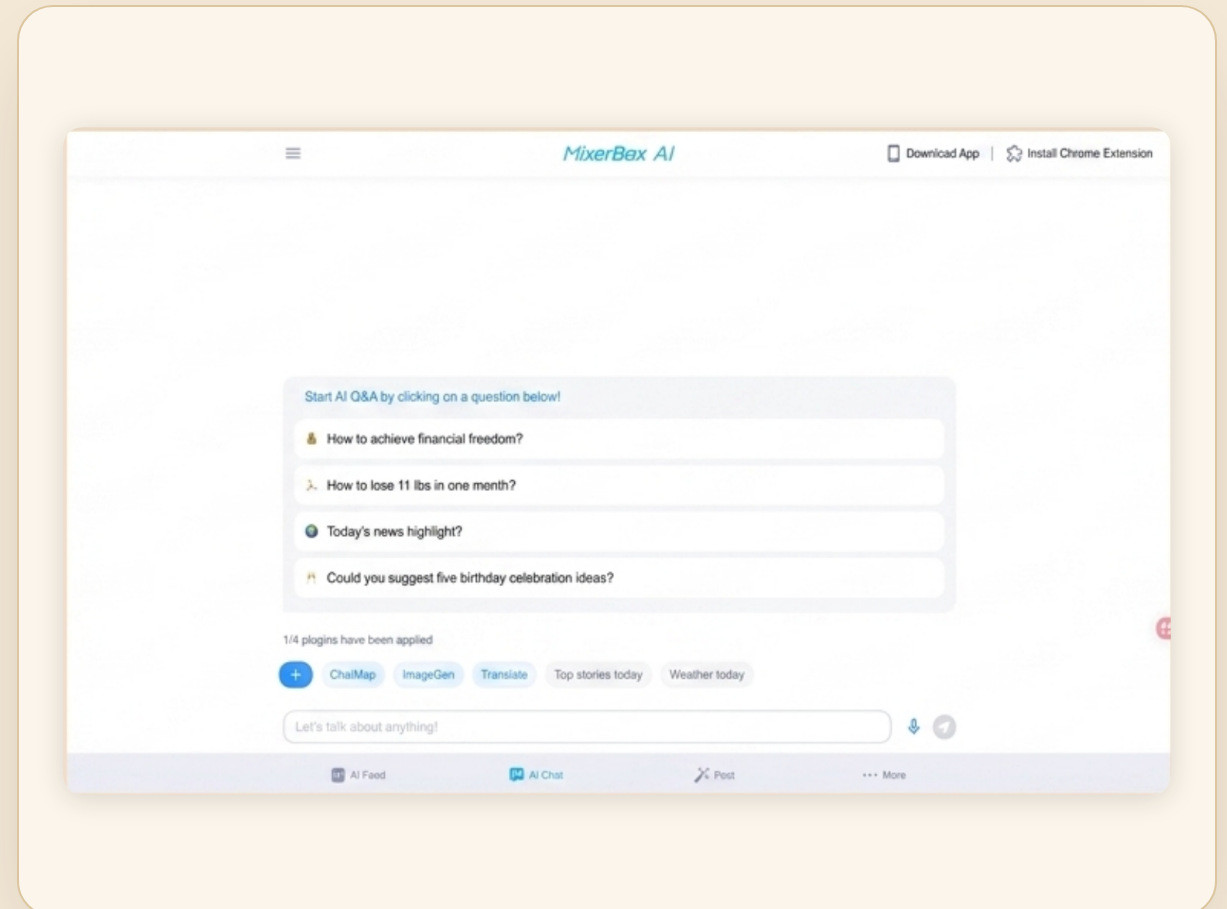
Engineering robust fallbacks for **LLM hallucinations, API timeouts, and context failures** at massive scale.

Impact

Reliably served **2.4M+** daily requests and handled **9,000** peak RPM.

2.4M+ REQUESTS / DAY

9,000 PEAK RPM



PROJECT 04

MixerBox Pay

Instant Checkout & Payment-Link Platform

Project Goal

Engineer a digital commerce checkout workflow capable of handling **sudden flash sales**.

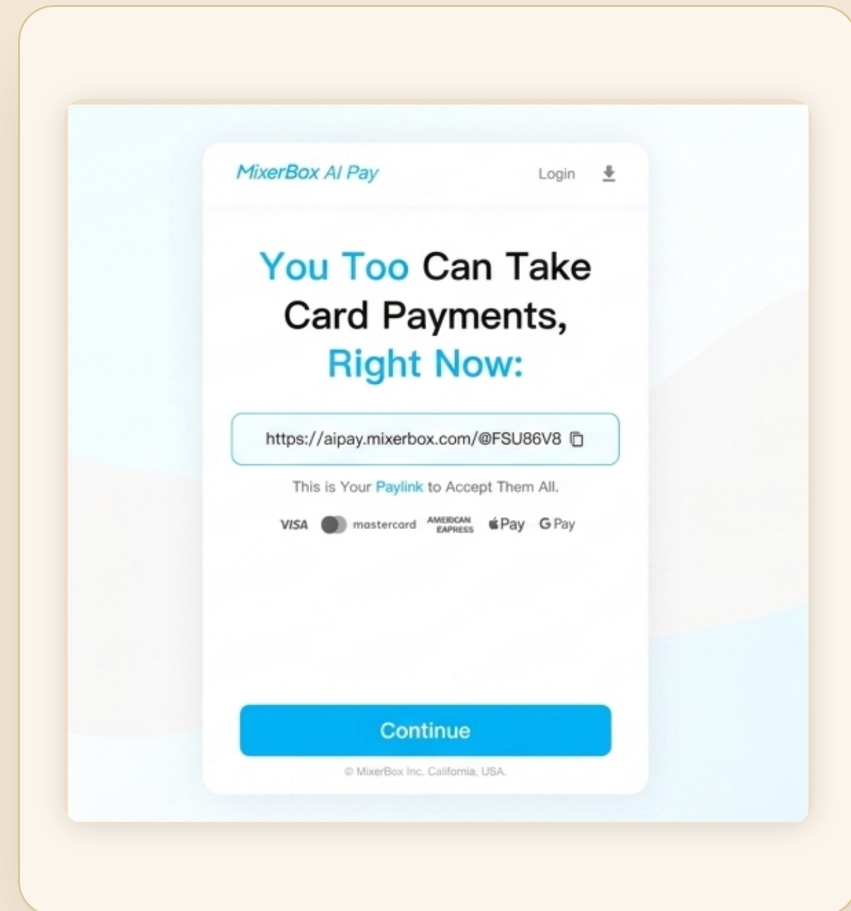
Core Challenge

Using **DynamoDB Transactions** to maintain strict inventory consistency and prevent overselling during massive traffic spikes.

Impact

Ensured **100%** transaction safety for thousands of concurrent users competing for limited stock.

100% TRANSACTION SAFETY



PROJECT 05

MixerBox ID

Centralized Authentication & Identity Service

Project Goal

Build a centralized authentication, identity, and user-management service.

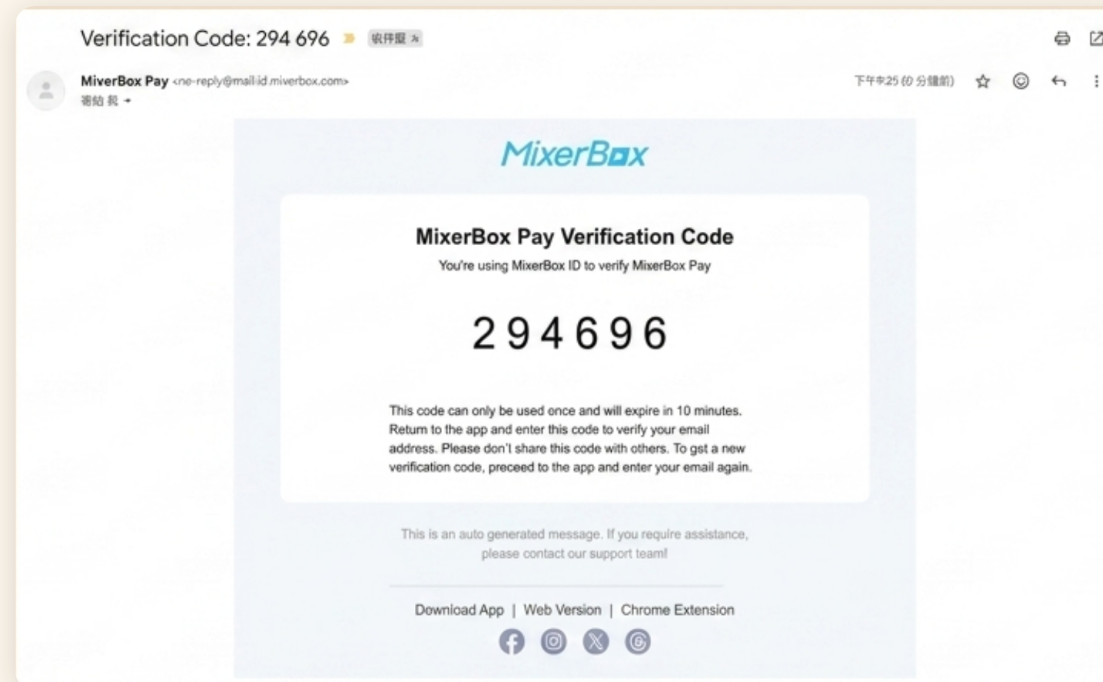
Core Challenge

Designing scalable, secure, and universally applicable **identity infrastructure** across a diverse product ecosystem.

Impact

Successfully authenticated and managed data for over **1,000,000** active users.

1M+ ACTIVE USERS



How I Build

Design for Unpredictability

Building deterministic boundaries around non-deterministic AI models.

Architect for Concurrency

Prioritizing strict transactional consistency during high-velocity traffic spikes.

Optimize for Reliability

Implementing robust error handling, fallbacks, and real-time synchronizations at scale.

CONTACT

Let's Build Something Together

I'm always happy to talk shop — new roles, interesting problems, or just AI systems and backend architecture.

Reach out anytime.

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